

PumpRupee

Find the rupees your pumps waste, before you buy a VFD

No flow meter

No shutdown

₹ waste by cause

Team ANS_4X

Schneider Electric Yuva Yodha Energy Tech Hackathon 2026

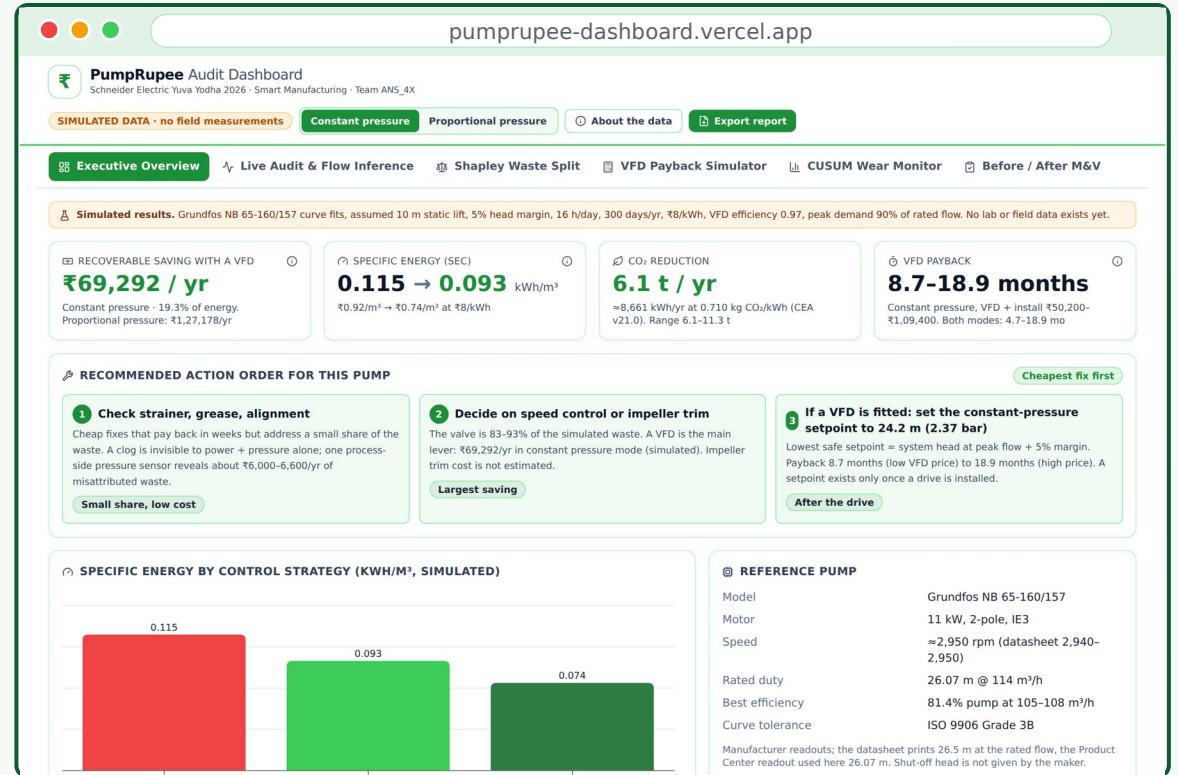
Track: Smart Manufacturing · Electrify · Automate · Digitalise



Live prototype dashboard

<https://pumprupee-dashboard.vercel.app/>

Simulated data · scan the QR code

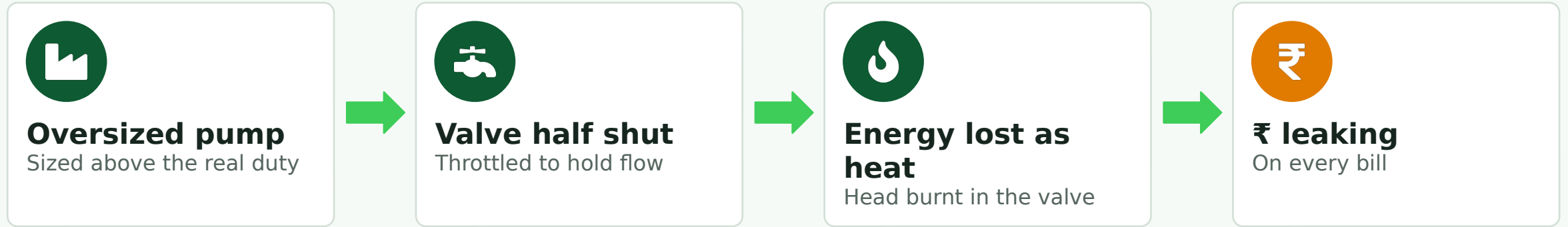


Working prototype dashboard · simulated data

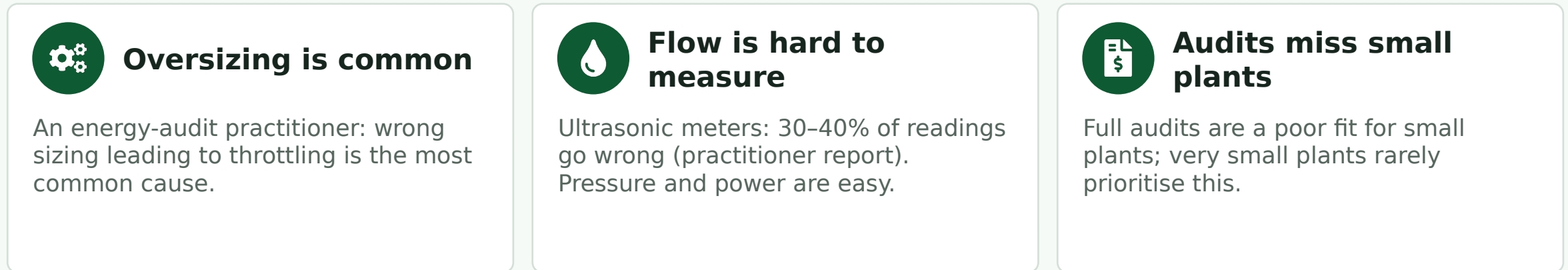
All results are simulated (Grundfos NB 65-160/157, 11 kW). Prices are indicative, not quotes.

Problem Statement

Oversized pumps run throttled. The extra energy is burnt across a valve, every hour, and nobody sees it.



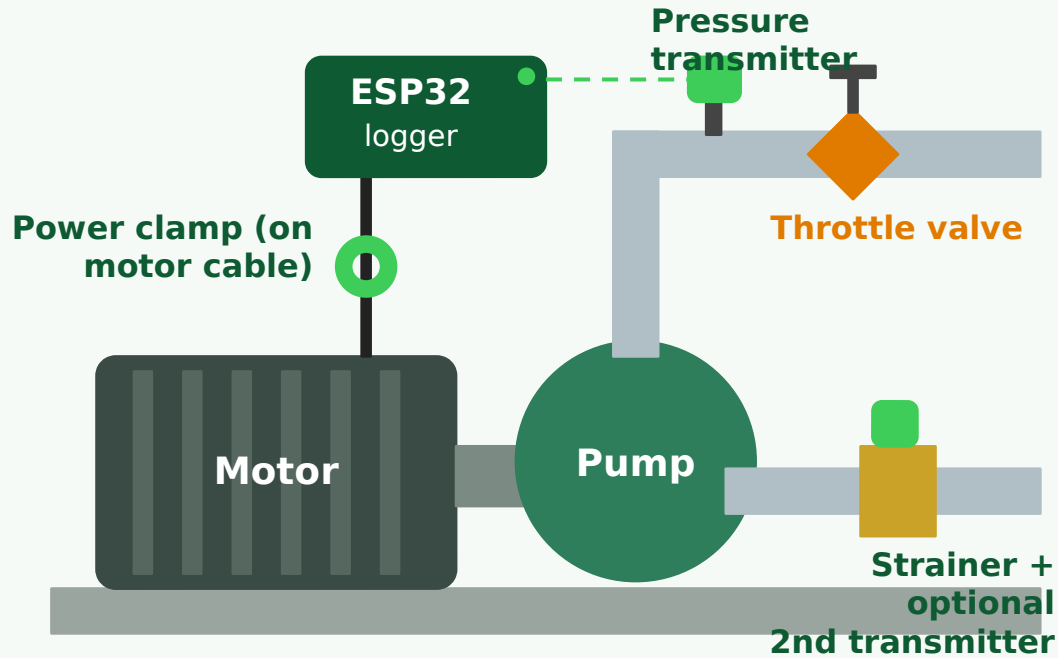
You pay for the pump AND for fighting the valve.



Head (pressure) and electrical power are cheap to measure. Flow is not. PumpRupee infers flow from the two that are.

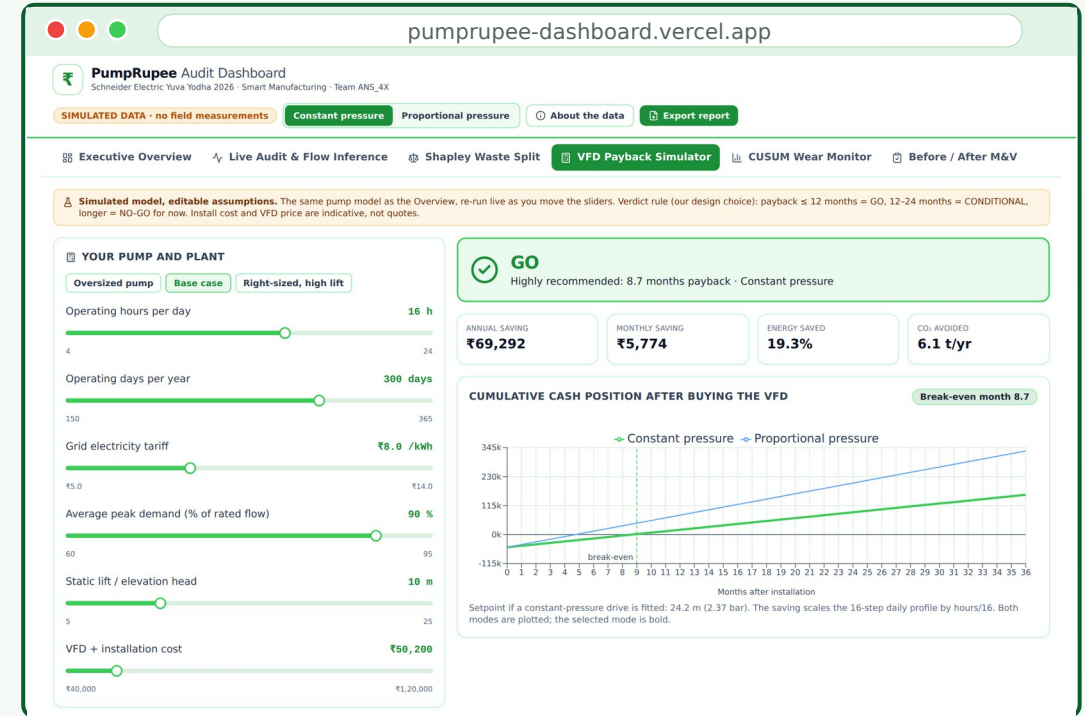
Proposed Solution

A clamp-on kit and a web dashboard turn power + pressure into “₹ wasted per year, and why”.



Concept illustration of the kit on a pump skid

Clamp on → log 7-14 days → ₹ report → the right fix



Working prototype dashboard · simulated data

- ✓ ₹ waste per year, by cause
- ✓ VFD payback verdict for THIS pump
- ✓ Cheapest fix first + safe setpoint
- ✓ Verified saving after the fix

Audits and recommends. Does not control the pump.

Alignment to Smart Manufacturing

Electrify · Automate · Digitalise, at an entry cost an MSME can afford.



Electrify

- Makes motor energy visible with a clamp-on power measurement
- Points owners to the pumps where a right-sized drive pays back



Automate

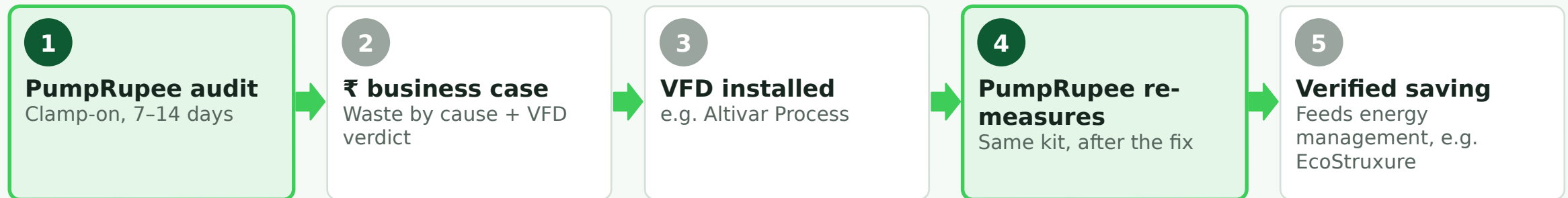
- Automatic diagnosis of waste by cause
- Wear-drift alarm (CUSUM) on the pump
- Lowest safe VFD setpoint; the drive does the controlling



Digitalise

- Logs become a ₹ business case on a dashboard
- Before/after proof of the saving
- Owner keeps the data and can use any drive

How PumpRuppee feeds drives and energy management



A low-cost front door that expands VFD and energy-management adoption into MSMEs that skip audits today. Concept only: no drive or platform integration built.

Key Features & User Journey

Seven steps from clamp-on to a verified saving.

-  **Clamp on** Power clamp on the motor cable
-  **Fit sensors** Pressure on existing gauge ports
-  **Log 7-14 days** ESP32 saves a CSV of power + pressure
-  **Upload to the dashboard** ₹ report by cause
-  **Apply the fix** Cheapest first; VFD verdict
-  **Re-measure** Verify the saving
-  **Move the kit** Next pump

₹ waste by cause

No flow meter

No shutdown

VFD verdict

Upload your logs

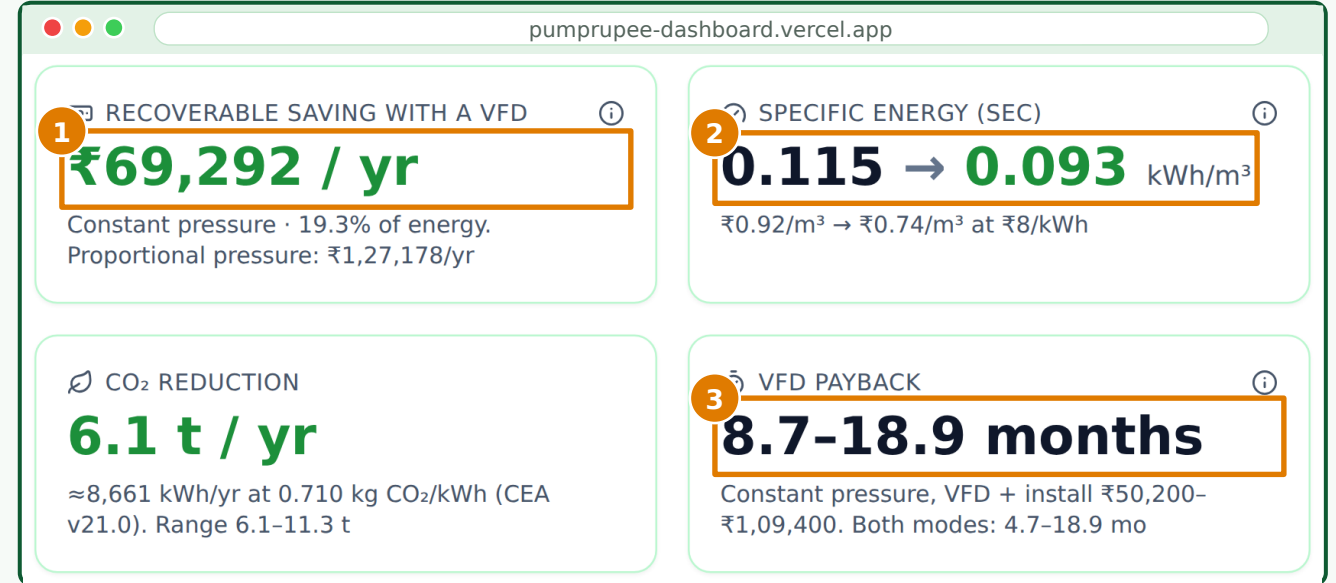
Reusable kit



Live prototype dashboard

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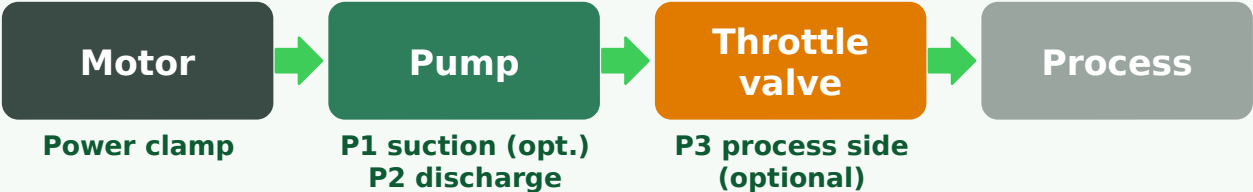
Working prototype dashboard · simulated data

- 1 ₹ saving per year with a VFD (simulated)
- 2 Specific energy: before → after (kWh/m³)
- 3 VFD payback range at ₹50,200-1,09,400

New: upload your own logs

Load a CSV of power and pressure; the dashboard calibrates the pump in your browser and prices the saving (slide 8).

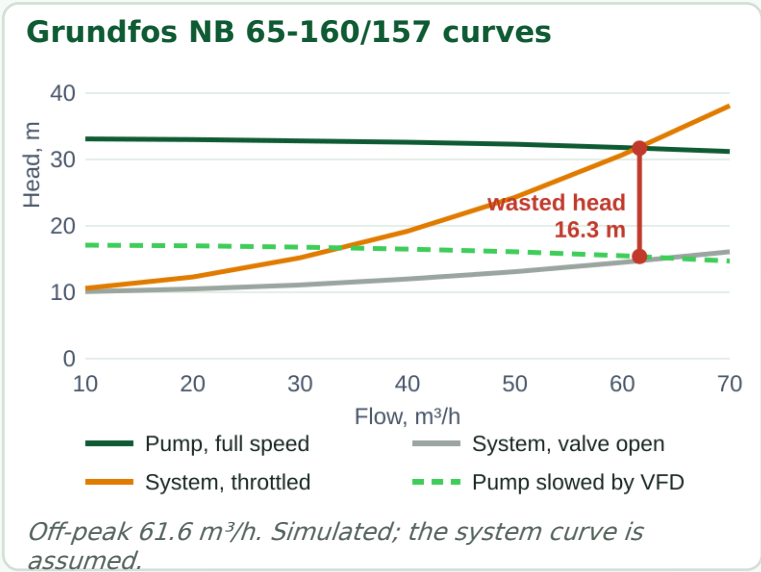
Technical Approach



Minimum kit: clamp + P2. Add P3 before a VFD purchase.

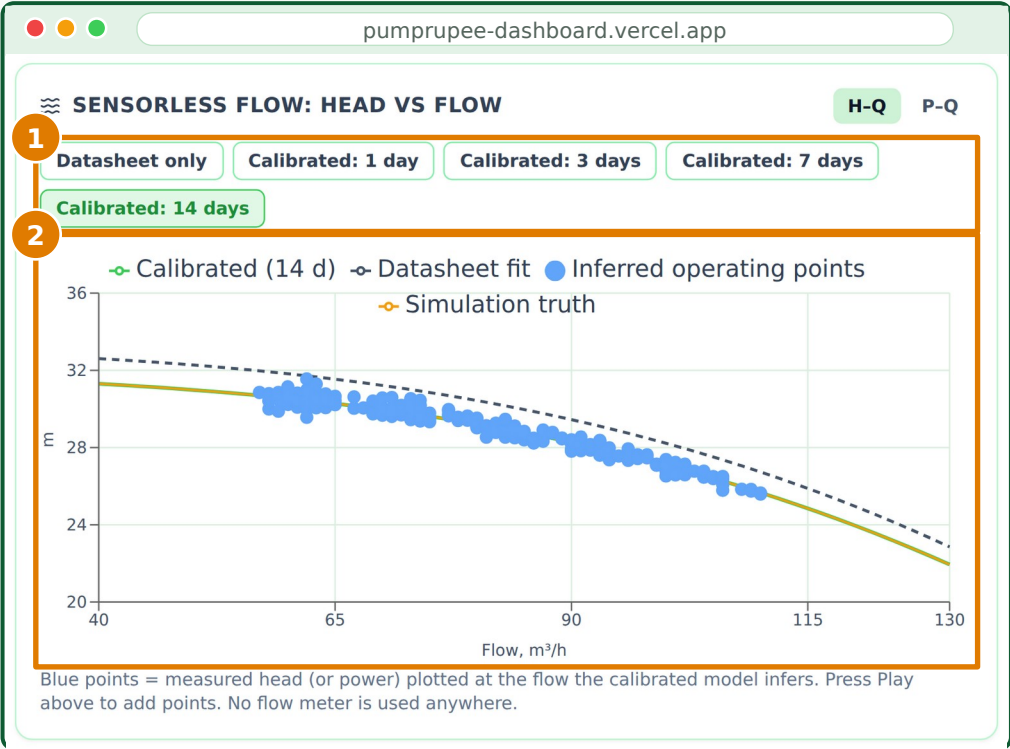


Shapley = fair split of blame between causes. **CUSUM** = early alarm when the pump slowly wears. **SEC** = electricity units per m^3 pumped.



How flow is inferred

- Head $\rightarrow$ flow from the pump curve $H(Q)$
- Power $\rightarrow$ shaft power via motor efficiency $\rightarrow$ flow
- Two scales, s_h and s_p , fitted from 7–14 days of logs
- Optional anchors: shut-off head, closed-valve power
- No flow meter used



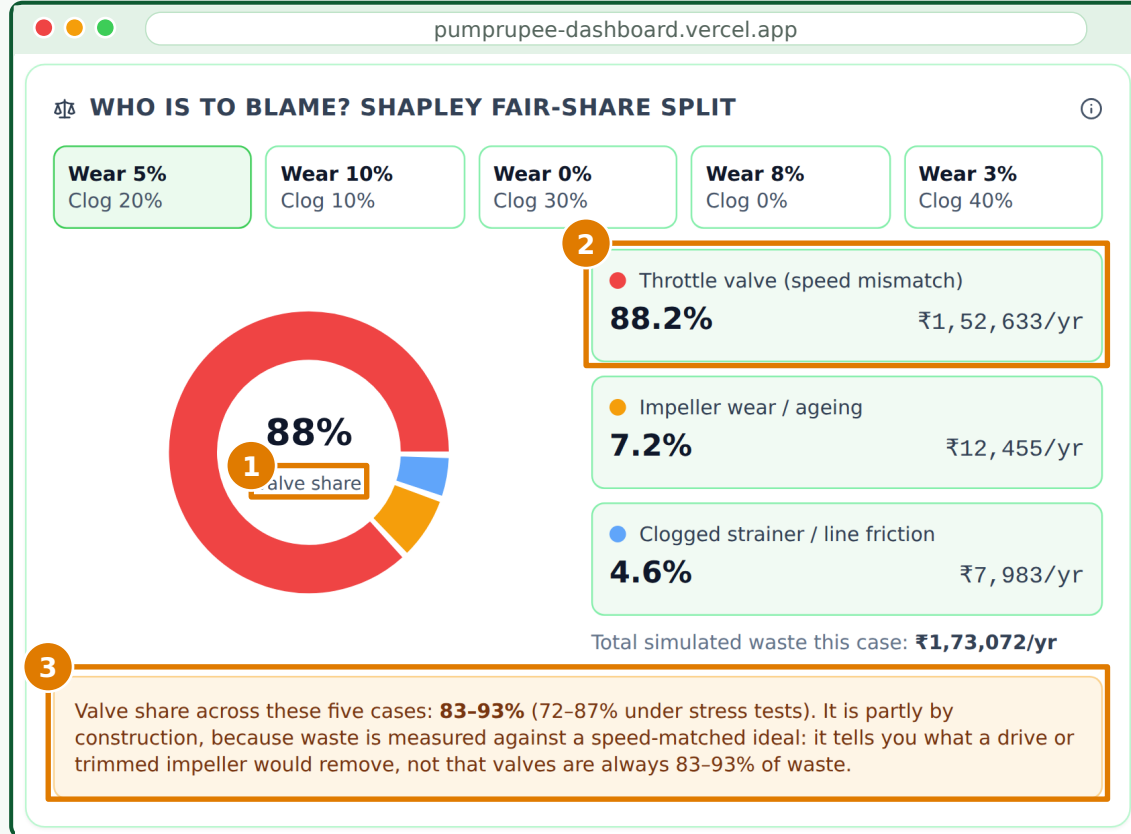
Working prototype dashboard · simulated data

- 1 Choose datasheet-only or 1/3/7/14 days of logs
- 2 Blue points: measured head at the flow the model infers (no flow meter)

About ± 3 points only with a measured system curve; otherwise about ± 10 .

Prototype & Simulated Results

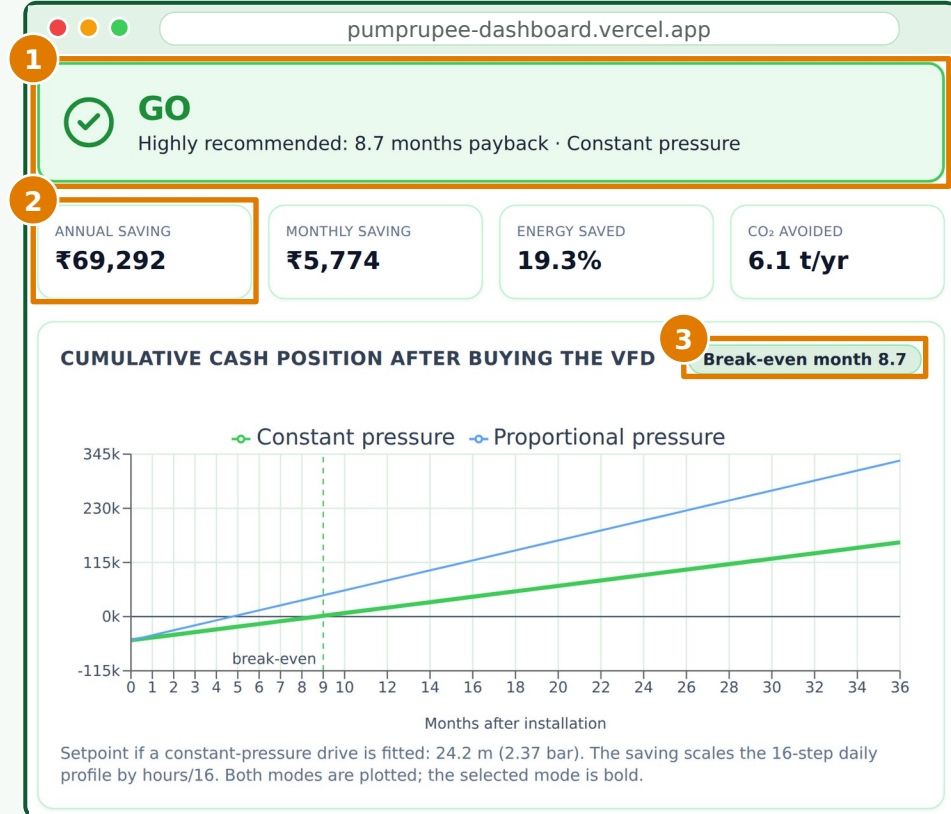
SIMULATED - Grundfos NB 65-160/157, 11 kW · 10 m lift · 5% margin · 16 h/day · 300 days · ₹8/kWh



Working prototype dashboard · simulated data

- 1** Valve share of waste (Shapley split)
- 2** ₹ per year, by cause
- 3** Valve share is partly by construction

₹69,292 / ₹1,27,178 per year · valve 83-93% of waste · payback 4-19 months · ±3 / ±10 points

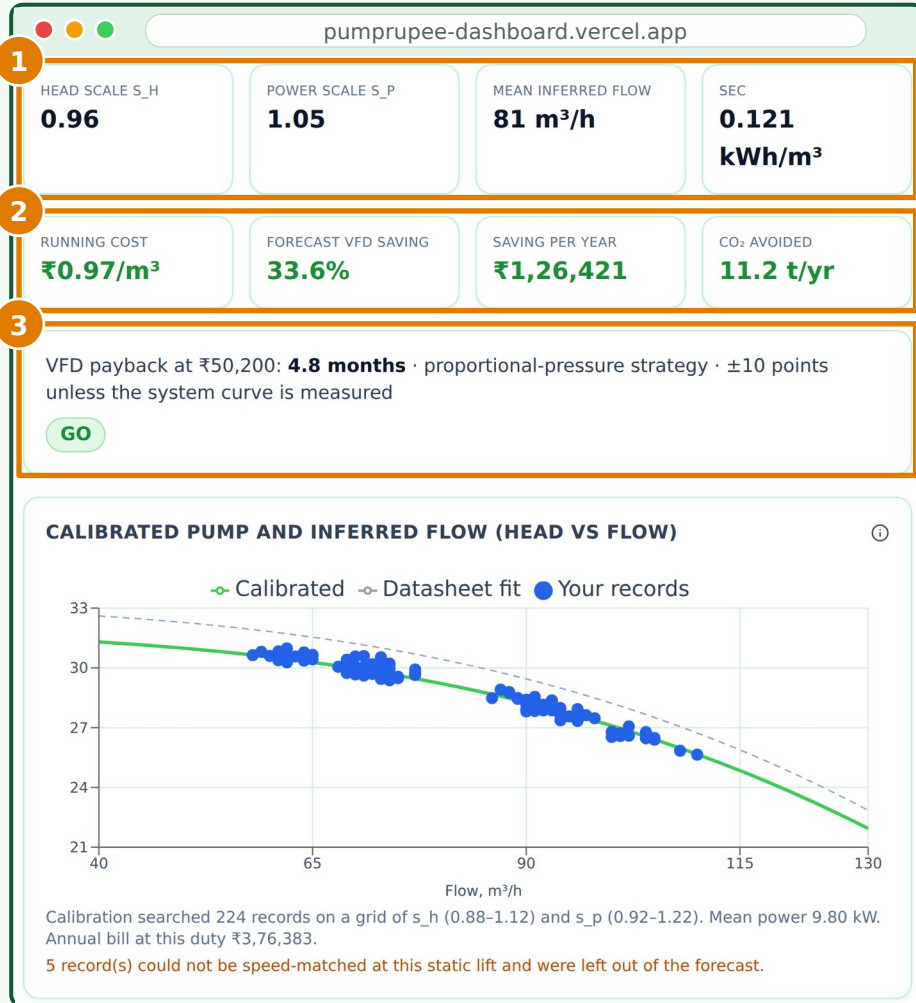


Working prototype dashboard · simulated data

- 1** GO / NO-GO verdict for this pump
- 2** Annual saving in ₹
- 3** Break-even month

Prototype: Upload Your Own Logs

Not just a replay: the dashboard calibrates from a CSV you load.



Working prototype · sample data



Export a CSV

power_kw, discharge_bar and optional suction_bar, from the ESP32 SD card or any logger.



Stays in your browser

Nothing is uploaded to a server.



Calibrate and infer flow

Fits a head scale and a power scale, then infers flow per record. No flow meter.



₹ saving and payback

SEC, ₹/m³, VFD saving per year and payback for your tariff and VFD price.

- 1 Calibrated pump scales found from the logs
- 2 Forecast VFD saving, running cost, CO₂
- 3 VFD payback and GO / NO-GO

Limits: reference pump (Grundfos NB 65-160/157) only; system curve assumed, so about ±10 points unless measured.

Innovation

We build on proven tools. The new part is the integration for MSMEs.

Already exists (we build on it)

- KSB PumpMeter, US DOE PSAT
- Samotics (motor-current efficiency)
- Schneider Altivar Process (sensorless flow), EcoStruxure
- Indian IoT vendors and ESCOs
- Shapley split and CUSUM: known methods

Our integration

- ₹ waste split by cause
- Prices the value of an extra sensor
- Low-cost shared kit, no flow meter
- Built-in before/after verification
- Pump-specific VFD verdict before capex

	Flow meter	Manual audit	Full platform	PumpRupee
Upfront cost, small plant	Meter + install	Fee per visit	Higher	₹6,100-13,300
No shutdown	—	✓	—	✓
Answer in ₹	✗	✓	—	✓
Cause breakdown	✗	✓	—	✓
VFD verdict	✗	✓	—	✓
Verified savings	—	—	✓	✓
Fits small MSMEs	—	✗	—	✓

✓ yes ~ partly / varies ✗ no. Our general assessment; PumpRupee's ticks are design targets (simulated).

We found no low-cost shared kit that combines these. Full platforms are the next step after PumpRupee.

Expected Impact

SIMULATED - per 11 kW pump; prices indicative, not quotes

A one-time kit vs a yearly saving (₹)

Kit, one-time and reusable (top tier)

 **₹13,300**

Saving per year, constant pressure

 **₹69,292**

Saving per year, proportional pressure

 **₹1,27,178**

One kit ≈ 5-19% of a year's simulated saving on one pump.



6.1-11.3 t

CO₂ avoided per pump per year (0.710 kg/kWh)



₹50,200-1,09,400

wrong VFD purchase avoided where it won't pay back



4-19 months

payback, oversized pump (2-4+ years if right-sized, high lift)

Who benefits

MSME plant owners

Maintenance teams

Energy auditors / ESCOs

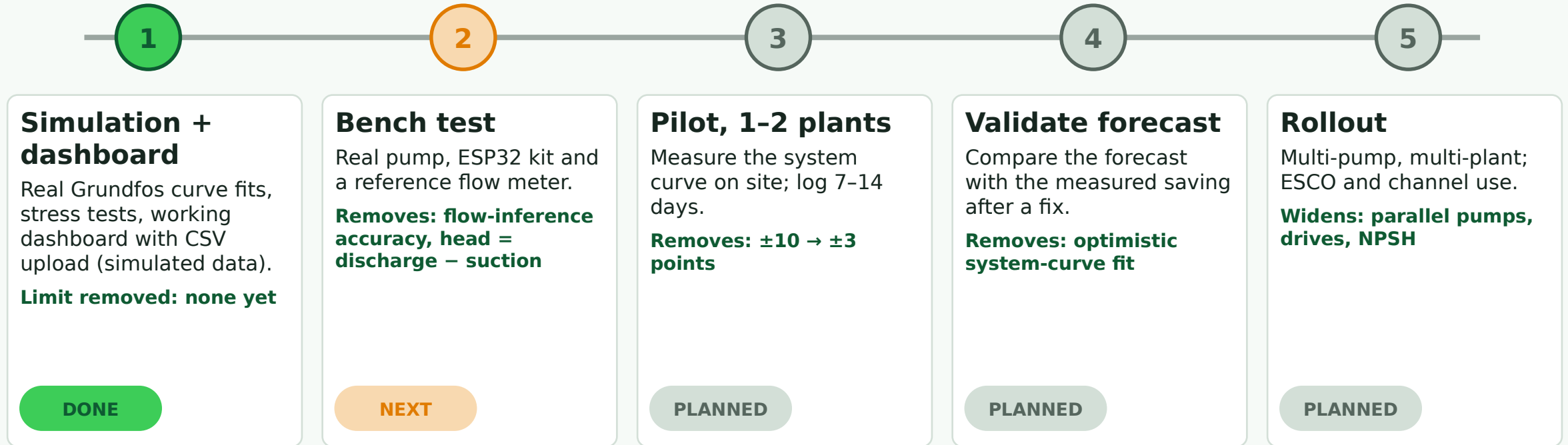
Drive channel partners

India's industrial efficiency goals

Scale-up illustration, not a projection: total = N pumps × per-pump saving. For N = 10: ₹6,92,920-12,71,780 per year and 61-113 t CO₂ per year, using the 11 kW base case. The 54-60% saving for heavily oversized pumps is an upper bound only.

Implementation Roadmap

What is done, what is next, and which limit each phase removes.



Honest limits we validate next

- Simulation and software prototype only: no lab or field data, no real quotes
- About ± 3 points only with a measured system curve; otherwise about ± 10 ; the curve fit is optimistic
- Single pump, steady state. Out of scope: parallel pumps, closed loops, pumps already on drives, min-flow/NPSH, control dynamics

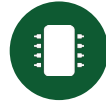
Team Introduction

Team ANS_4X · Schneider Electric Yuva Yodha 2026 · Smart Manufacturing



Akshaya V G
Team Lead &
Software/Dashboard

Data pipeline and the
PumpRupree web dashboard



Adithyaa J
Embedded Hardware

ESP32 logger, power clamp and
pressure-sensor integration



Nikilaesh B
Energy Analytics

Pump hydraulics model,
Shapley waste split and VFD
payback logic



**Siva
Subramaniam S**
Simulation & Validation

Pump-curve simulation, CUSUM
wear-drift testing, pilot and
roadmap planning

References & data sources

- Pump data: Grundfos Product Center duty-point readouts for NB 65-160/157 (product no. 97839240), retrieved 2026-10-02. Curve tolerance ISO 9906:2012 Grade 3B. Curves in this deck are cubic fits to those 14 points.
- Emission factor: CEA CO₂ Baseline Database for the Indian Power Sector, v21.0 (FY2024-25): 0.710 kg CO₂/kWh.
- Energy-audit practitioner input (paraphrased, not named): wrong sizing is the most common cause of throttling; 30–40% of ultrasonic flow readings go wrong.
- Method: affinity laws; Shapley value attribution; CUSUM change detection; IPMVP-style measurement and verification.
- Component prices (indicative online listings, 2026-10-02, not quotes): ESP32 (Robu); SCT-013 clamp (ElectronicsComp, Robodo, Robocraze); ZMPT101B voltage sensor; 4–20 mA pressure transmitters (Nishka, Utopia, Shri Instruments); VFDs (Delta MS300 listing, Industrybuying). Install cost ₹15,000 is an assumption.
- Prior art: KSB PumpMeter, US DOE PSAT, Samotics, Schneider Altivar Process and EcoStruxure, Indian IoT vendors and ESCOs.
- Simulation: our open Python scripts (fixed seeds) in the project repository, folder yuva_yodha/; dashboard source in yuva_yodha/dashboard/.
- All results are simulated. No lab or field data.



Live prototype dashboard
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**Cheap kit. Clever inference. Answers
in rupees.**